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CMSC 207 CRN 32950

CMSC 207: Chapter 6 Homework

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1. Sets A and B are defined as follows:

$$A = \{n \in \mathbb{Z} \mid n = 8r - 3 \text{ for some integer } r\}$$
$$B = \{m \in \mathbb{Z} \mid m = 4s + 1 \text{ for some integer } s\}.$$

   a. Prove that $A \subseteq B$.
   b. Disprove that $B \subseteq A$.

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a.

b.

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2. Prove by contradiction that for all sets A and B, $(A - B) \cap B = \emptyset$.
